

Algorithms Problem Solving

Level 3

Add Clients to File



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Definition

Interactive Batch Client Entry

-  Program allows the user to enter multiple client records during one execution.
-  Each client is saved to the file before the program asks whether another client should be added.

Explanation

Repetition Control

-  A `do-while` loop is suitable because at least one client entry should occur before asking whether to continue.

-  The loop repeats based on the user's `Y/N` response.

Client Creation

-  `ReadNewClient()` collects the fields required to create one client record.

-  The resulting structure is passed to the file-saving function.

Persistent Storage

-  `SaveClientDataToFile()` serializes and appends the current client.

-  The process is repeated for every additional client.

Termination Decision

-  After each successful entry, the user is asked whether another record should be added.

-  The loop terminates when the response is not a continuation response.

Procedures

1. Start the client-entry loop.
2. Call `ReadNewClient()`.
3. Save the new client to the file.
4. Ask `Do you want to add more clients? (Y/N)`.
5. Continue for `Y` or `y`.
6. Stop for `N` or `n`.

💡 Example

Client 1 saved successfully.

Do you want to add more clients? Y

Client 2 saved successfully.

Do you want to add more clients? N

➡ Result

A150#/#1234#/#Mohammed#/#079#/#5000

Note

- ◆ A loop is more appropriate than recursion for repeated user input.
- ◆ Each client is saved independently.
- ◆ The program can continue adding records until the user chooses to stop.

⊘ Warning

- ◆ Do not recursively call `AddNewClient()` to implement an indefinite input loop.
- ◆ Input-buffer handling is important when switching between numeric/character input and `getline()`.
- ◆ The program should validate the Y/N response rather than assuming every other character means "No".



Comparison

▨ `AddNewClients()` Loop

- ✓ Uses controlled iteration for repeated data entry.
- ✓ Keeps memory usage predictable.
- ✓ Allows many records to be entered in one session.

▤ Single Client Addition

- ✓ Adds only one record per program execution.
- ✓ Is simpler but requires restarting the application for another record.
- ✓ Does not provide batch entry.

✦✦ Summary

- ✓ Uses a loop to support repeated client creation.
- ✓ Reads and saves each client record independently.
- ✓ Uses a user-controlled continuation condition.
- ✓ Demonstrates practical interactive data-ingestion flow.

Thank you 😊